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Cosmos Unveiled: Dark Matter Halos

The Invisible Scaffolding of Galaxies

Aira Astralis

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Introduction

Every galaxy that astronomers can see is wrapped in a far larger one that they cannot. The visible spiral, the disc of stars and gas that fills a telescope's field of view, sits at the center of an enormous, roughly spherical, nearly invisible mass of something we have never directly detected. That unseen envelope is the dark matter halo, and in modern cosmology it is not a peripheral correction to the picture of a galaxy but the main structure of the galaxy itself. The stars, the gas, the dust, the planets, and every living thing are a thin luminous frosting spread across the top of a vast invisible cake.

A dark matter halo is defined by its behavior rather than by its appearance, because it has no appearance in the ordinary sense. It is a region of space that has decoupled from the expansion of the universe and collapsed under its own gravity into a bound system. Within that region, matter orbits a common center, and the whole assembly satisfies the virial theorem: the kinetic energy of its internal motions balances its gravitational binding energy. A single halo may contain many smaller bound clumps within it, called subhalos, the gravitational fossils of the smaller structures it has absorbed over cosmic time. In the standard model of cosmology, called Lambda-CDM, halos are the fundamental units of structure. Galaxies are not the primary objects of the universe; halos are, and galaxies form inside them.

The halo of a galaxy like the Milky Way extends far beyond the edge of the visible disc. Where the starlight fades out at tens of thousands of light years from the center, the halo keeps going, thinning gradually for hundreds of thousands of light years, enveloping the entire visible structure the way an ocean envelops a floating cork. Its total mass outweighs the visible galaxy several times over. Nothing in it shines, reflects, absorbs, or emits any light that any telescope has ever gathered, and yet its presence is not in doubt, because its gravity is felt everywhere: in the orbital speeds of stars, in the paths of gas clouds, in the bending of light from objects behind it, and in the motions of entire galaxies around one another.

The evidence that first forced this conclusion on astronomy was deceptively simple. In our own solar system, the planets closest to the Sun move fastest, and the outer planets crawl, with orbital speeds that fall as the square root of distance, exactly as Kepler's third law and Newton's gravity require when nearly all the mass sits at the center. If a galaxy worked the same way, with its mass concentrated in its luminous bulge, its stars should orbit in the same slowing pattern. Instead, when astronomers finally measured the orbital speeds of stars and gas at ever greater distances from the centers of spiral galaxies, they found something else: the speeds refuse to fall. Out at the faint edge of the disc, where almost no light shines, stars and hydrogen clouds orbit as fast as, or faster than, the stars in the bright interior. Something with mass, a great deal of mass, is present where nothing can be seen.

This book is the story of that invisible architecture. It follows the long chain of measurement that led astronomers, one reluctant step at a time, from the first anomalous velocities of the Andromeda galaxy to the modern consensus that roughly five sixths of all matter in the universe is dark. It explains what a halo is, how halos form in the early universe from the gravitational collapse of cold dark matter, and how they grow over billions of years by merging with one another in a hierarchy that runs from the smallest clumps to the great halos that hold clusters of galaxies. It describes the ways halos are measured: rotation curves traced with radio telescopes, gravitational lensing that maps mass directly, X-ray halos of hot gas, and the delicate census of stellar streams left behind by devoured dwarf galaxies. And it follows the search for what the dark stuff actually is, a search that has eliminated candidate after candidate and continues today in detectors buried deep underground, in particle colliders, and in the patterns of the cosmic microwave background.

The dark matter halo is, in a real sense, the scaffolding of the universe. The ripples in the early universe became the seeds of halos, the halos became the sites where gas collected and cooled and lit itself as the first stars, and those first galaxies assembled through halo mergers into the galaxies we see around us today. To understand halos is to understand how structure in the universe is built. That is the journey this book takes, and it begins with the first person who noticed that a galaxy was not behaving as it should.

Chapter 01: The Discovery of Dark Matter Halos

The discovery of dark matter halos did not begin with dark matter at all. It began with a disagreement about speed. In 1914, the American astronomer Vesto Slipher was making spectroscopic observations of the Andromeda galaxy, the great spiral nearest to our own, and he noticed something curious in the light of its central bulge. The stars on one side of the bulge were approaching him at about 320 kilometers per second, while the stars on the

other side were receding at about 280. The difference was the signature of rotation: the galaxy's disc was turning, spinning one edge toward us and the other away, and Slipher's measurement was the first evidence that galaxies rotate at all. It was a landmark observation, and like many landmark observations, the thing it would eventually imply was far stranger than the thing it directly showed.

Slipher's instrument could only sample the bright central bulge, where the light of billions of stars blends into a single smear. To trace rotation across the whole disc, astronomers needed to measure the speeds of individual regions one at a time. In 1918, Francis Pease carried that work further, determining the rotation speed within the central region of Andromeda, and his results indicated that the inner part of the galaxy rotates at a roughly constant angular speed, like a wheel. That finding sat comfortably within expectations. A compact, rigidly turning core suggested that the galaxy's mass and light were concentrated together, and that farther out the orbital speeds would duly decline, planet-like, as the gravitational grip of the central mass weakened. Nature, it would turn out, had other plans.

The next voice in the story belonged to Jan Oort, the great Dutch astronomer of the Milky Way. In 1932, Oort studied the motions of stars in the solar neighborhood and reported that they moved faster than expected if one assumed the mass was distributed according to the visible light. He was, in effect, weighing the disc of the Milky Way near the Sun and finding the scale reading too high. These particular measurements were later shown to be essentially erroneous, a rare wrong step for so careful an astronomer, but the method he used, comparing the dynamics of nearby stars to the light they belong to, was sound, and it would return with better data decades later. Oort also worked on the internal motions of the Milky Way using quantities now called the Oort constants, and he pursued the problem of galactic rotation with the Carnegie Double Astrograph, an instrument built expressly for the problem. The machinery for weighing galaxies was being assembled, even before anyone knew it would need to be used.

Meanwhile, in 1933, a far more dramatic argument was unfolding in Switzerland. Fritz Zwicky, a flamboyant and famously pugnacious astronomer at Caltech, applied the virial theorem to the Coma cluster of galaxies. The theorem relates the average speed of a gravitationally bound system's members to the total mass holding them together. Zwicky measured the speeds of the Coma galaxies and found them far too fast for the mass implied by the visible stars. Either the cluster was flying apart, or it contained an immense quantity of unseen mass. He coined a name for it, dunkle Materie, dark matter, and estimated that the cluster held hundreds of times more of it than the luminous matter suggested. His numbers were too large, for reasons of measurement and method, but the core of his reasoning was correct, and the term he coined has outlived every objection raised against it. Astronomers of the 1930s filed the result away as a curiosity; the strange case of the Coma cluster would wait four decades for its explanation.

The decisive confrontation came in Andromeda. In 1939, Horace Babcock, in his doctoral thesis at Harvard, measured the rotation curve of Andromeda, the plot of orbital speed against distance from the center, and found that the ratio of mass to light increased markedly toward the outer parts of the galaxy. There was, in the outskirts, more mass than light could account for. Babcock, like nearly everyone before him, did not reach for missing matter. He attributed the anomaly to the absorption of light within the galaxy, or to some modification of dynamics in the outer disc, and his measurements themselves would later disagree with better ones. The pattern of the whole history is visible here: each observer found something unexpected, and each reached for a conventional explanation, because the alternative, that galaxies are dominated by matter no one can see, was simply too strange.

Radio astronomy finally forced the issue, and it did so with the most abundant atom in the universe. Neutral hydrogen gas emits a faint spectral line at a wavelength of 21 centimeters, a whisper of radiation predicted in the 1940s and first detected in 1951, and that whisper penetrates the dust that blocks visible light. Hydrogen gas extends far beyond the starlight in a spiral galaxy, tracing the disc to faint outer edges where no stars are left to measure. In 1957, Henk van de Hulst and collaborators, using the newly commissioned 25-meter radio telescope at Dwingeloo in the Netherlands, measured the rotation of M31, the Andromeda galaxy, in this hydrogen line, producing the first extended rotation curve in good agreement with modern data. A companion analysis by Maarten Schmidt showed that this curve could only be fit by a flattened mass distribution far more extensive than the light. In 1959, Louise Volders used the same telescope to show that the spiral galaxy M33 also failed to spin down as Keplerian dynamics demanded. The flat rotations were no longer a flicker in one galaxy's data; they were a pattern.

The capstone came from Ken Freeman in 1970, who examined the rotation curves of NGC 300 and M33 and noticed explicitly that the expected decline in velocity was simply not present in either galaxy, and considered

seriously that an undetected mass component must be responsible. And in the late 1960s and early 1970s, Vera Rubin, at the Carnegie Institution's Department of Terrestrial Magnetism, working with the sensitive spectrograph built by her colleague Kent Ford, measured the velocity curves of spiral galaxies to a greater degree of accuracy than had ever been achieved. In 1975, Rubin and Ford announced that the stars in spiral galaxies orbit at roughly the same speed across an enormous range of radii, which implied that the mass of a galaxy grows approximately linearly with distance from its center, far beyond the visible disc. Rubin's comprehensive paper of 1980 sealed the case with a sample of galaxies all showing the same behavior. The flat rotation curve had moved from anomaly to fact, and the dark matter halo, the enormous invisible mass it demanded, moved from speculation to the central problem of astrophysics. Half a century of hints had converged: the halos had been discovered by the simple, stubborn act of watching stars go around.

The discovery file also holds the argument from the clock, the field's oldest weighing of the invisible. The two great spirals of the local group, the Milky Way and its nearest giant neighbor, approach each other, and the approach has a arithmetic meaning: two galaxies born flying apart, slowed and reversed by their mutual pull, must be pulled by everything they have, and the observed slow closing speed of the pair weighs their combined mass at many times the count of their stars, the timing argument being the founding card of the halo case and one of the oldest numbers in the whole dark ledger, the card filing it as the argument's starting exhibit: even before any rotation curve was measured, the sky's own pendulum had already said the galaxies outweigh their light.